

SQL PROJECT- MUSIC STORE DATA ANALYSIS

Question Set 1

1. Who is the senior most employee based on job title?

The screenshot shows the SQL Developer interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
4  
5 /* Q1: Who is the senior most employee based on job title? */  
6  
7 • SELECT title, last_name, first_name  
8 FROM employee  
9 ORDER BY levels DESC  
10 LIMIT 1;  
11
```

The result grid shows the following data:

title	last_name	first_name
General Manager	Adams	Andrew

2. Which countries have the most Invoices?

The screenshot shows the SQL Developer interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
13 /* Q2: Which countries have the most Invoices? */  
14  
15 • SELECT COUNT(*) AS c, billing_country  
16 FROM invoice  
17 GROUP BY billing_country  
18 ORDER BY c DESC;  
19  
20
```

The result grid shows the following data:

c	billing_country
131	USA
76	Canada
61	Brazil
50	France
41	Germany
30	Czech Republic
29	Portugal
28	United Kingdom
21	India
13	Ireland

3. What are top 3 values of total invoice?

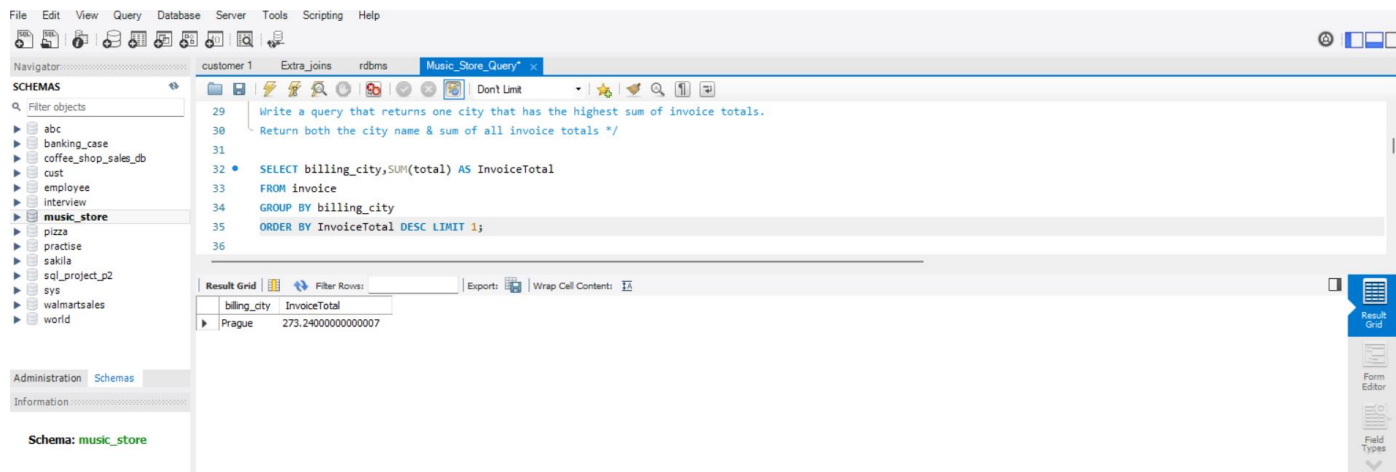
The screenshot shows the SQL Developer interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
19  
20  
21 /* Q3: What are top 3 values of total invoice? */  
22  
23 • SELECT total  
24 FROM invoice  
25 ORDER BY total DESC limit 3;  
26
```

The result grid shows the following data:

total
23.759999999999998
19.8
19.8

4. Which city has the best customers? We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals?



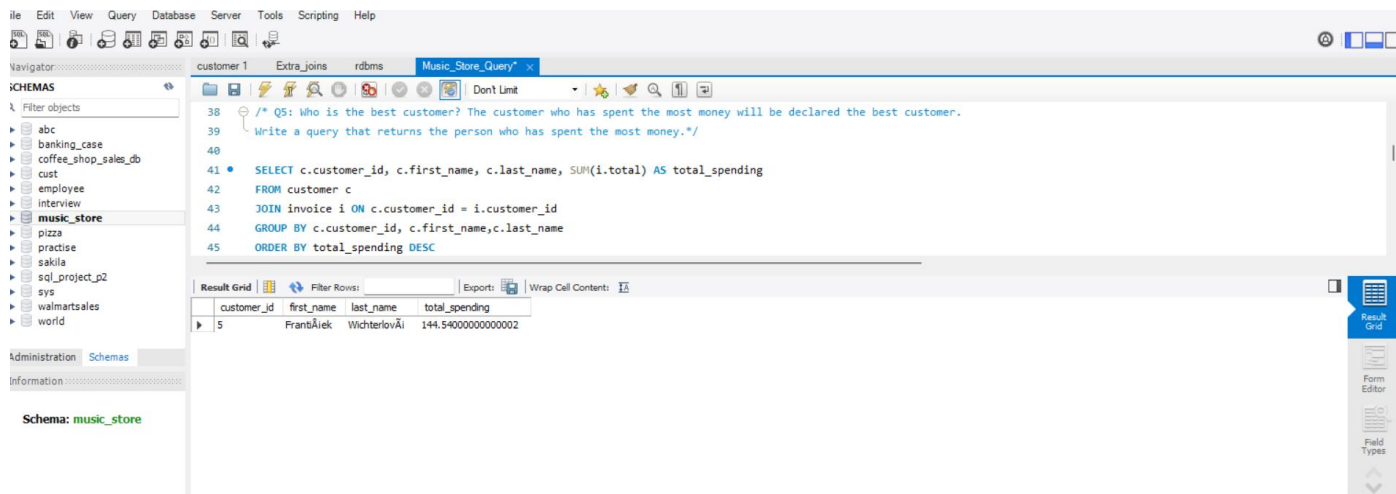
The screenshot shows the SQL Studio interface with the 'music_store' schema selected. The query editor contains the following SQL code:

```
29 Write a query that returns one city that has the highest sum of invoice totals.  
30 Return both the city name & sum of all invoice totals */  
31  
32 • SELECT billing_city, SUM(total) AS InvoiceTotal  
33 FROM invoice  
34 GROUP BY billing_city  
35 ORDER BY InvoiceTotal DESC LIMIT 1;  
36
```

The 'Result Grid' shows the following data:

billing_city	InvoiceTotal
Prague	273.240000000000007

5. Who is the best customer? The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money?



The screenshot shows the SQL Studio interface with the 'music_store' schema selected. The query editor contains the following SQL code:

```
38 /* Q5: Who is the best customer? The customer who has spent the most money will be declared the best customer.  
39 Write a query that returns the person who has spent the most money.*/  
40  
41 • SELECT c.customer_id, c.first_name, c.last_name, SUM(i.total) AS total_spending  
42 FROM customer c  
43 JOIN invoice i ON c.customer_id = i.customer_id  
44 GROUP BY c.customer_id, c.first_name, c.last_name  
45 ORDER BY total_spending DESC
```

The 'Result Grid' shows the following data:

customer_id	first_name	last_name	total_spending
5	František	Wichterlov	144.540000000000002

Question Set 2

1. Write query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A?

The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
53 SELECT DISTINCT email AS Email, first_name AS FirstName, last_name AS LastName, genre.name AS Name
54 FROM customer
55 JOIN invoice ON invoice.customer_id = customer.customer_id
56 JOIN invoice_line ON invoice_line.invoice_id = invoice.invoice_id
57 JOIN track ON track.track_id = invoice_line.track_id
58 JOIN genre ON genre.genre_id = track.genre_id
59 WHERE genre.name LIKE 'Rock'
60 ORDER BY email;
```

The result grid displays the following data:

Email	FirstName	LastName	Name
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	Björn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock
daan.peeters@apple.be	Daan	Peeters	Rock
diego.gutierrez@yahoo.ar	Diego	Gutiérrez	Rock
dmiller@comcast.com	Den	Miller	Rock
dominique.lefebvre@gmail.com	Dominique	Lefebvre	Rock
edfrancis@yahoo.ca	Edward	Francis	Rock
eduardo@woodstock.com.br	Eduardo	Martins	Rock
ella.sullivan@shaw.ca	Ella	Sullivan	Rock

2. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands?

The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
65
66 SELECT artist.artist_id, artist.name, COUNT(artist.artist_id) AS number_of_songs
67 FROM track
68 JOIN album ON album.album_id = track.album_id
69 JOIN artist ON artist.artist_id = album.artist_id
70 JOIN genre ON genre.genre_id = track.genre_id
71 WHERE genre.name LIKE 'Rock'
72 GROUP BY artist.artist_id, artist.name
```

The result grid displays the following data:

artist_id	name	number_of_songs
1	AC/DC	18
3	Aerosmith	15
8	Audioslave	14
22	Led Zeppelin	14
4	Alanis Morissette	13
5	Alice In Chains	12
23	Frank Zappa & Captain Beefheart	9
2	Accept	4

3. Return all the track names that have a song length longer than the average song length. Return the Name and Milliseconds for each track. Order by the song length with the longest songs listed first?

The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

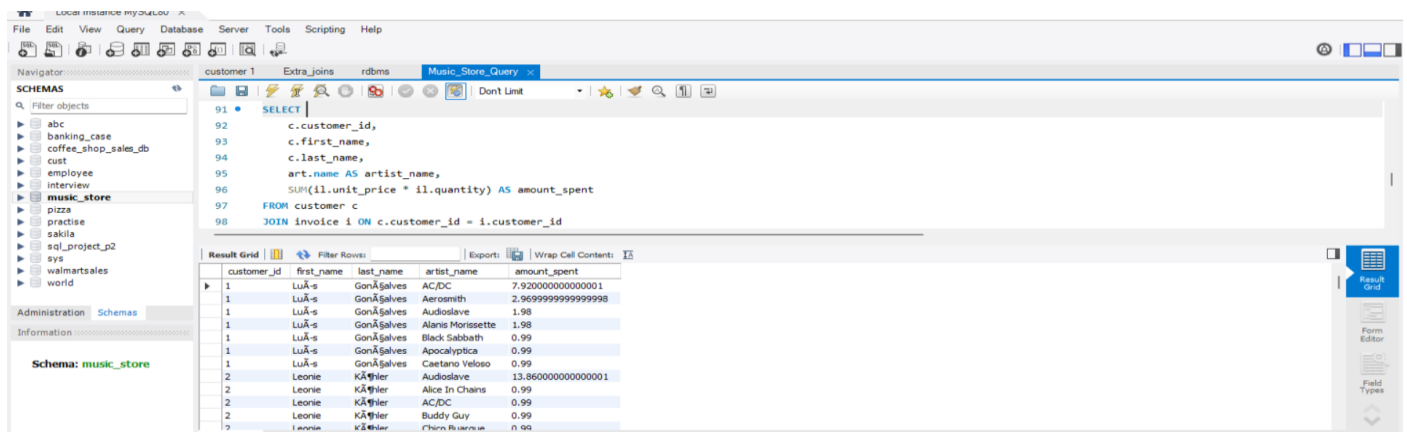
```
77 Return the Name and Milliseconds for each track. Order by the song length with the longest songs listed first. */
78
79 SELECT name, milliseconds
80 FROM track
81 WHERE milliseconds > (
82     SELECT AVG(milliseconds) AS avg_track_length
83     FROM track )
84 ORDER BY milliseconds DESC;
```

The result grid displays the following data:

name	milliseconds
How Many More Times	711836
Advance Romance	677694
Sleeping Village	644571
You Shook Me(2)	619467
Takin' Bout Women Obviously	589531
Stratus	582086
No More Tears	555075
The Alchemist	509413
Wheels Of Confusion / The Straightener	494524
Book Of Thel	494323
You Oughtta Know (Alternate)	491885
Terra	487420

Question Set 3

1. Find how much amount spent by each customer on artists? Write a query to return customer name, artist name and total spent?



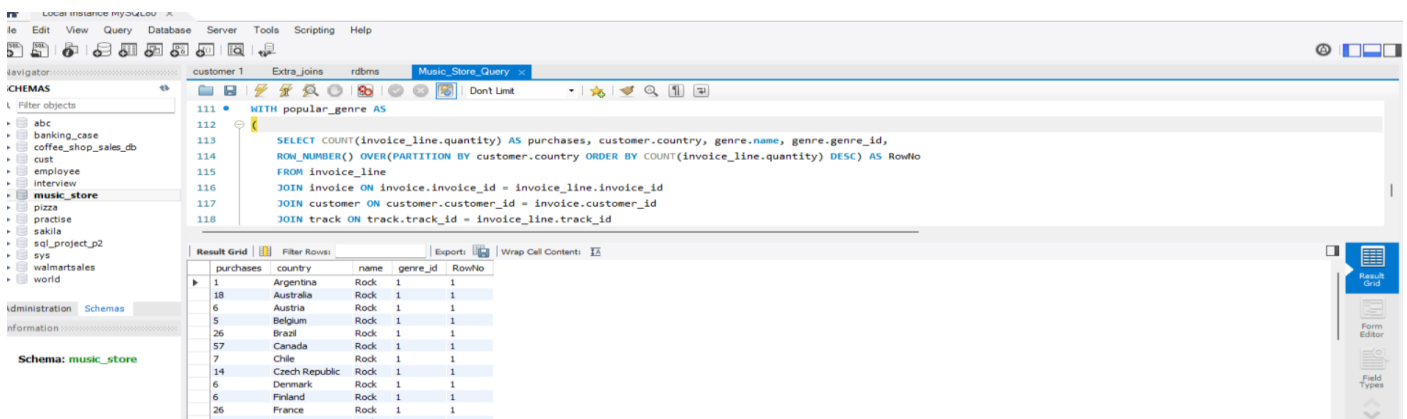
The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
SELECT
  c.customer_id,
  c.first_name,
  c.last_name,
  art.name AS artist_name,
  SUM(il.unit_price * il.quantity) AS amount_spent
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
```

The result grid displays the following data:

customer_id	first_name	last_name	artist_name	amount_spent
1	Luís	Gonçalves	AC/DC	7.920000000000001
1	Luís	Gonçalves	Aerosmith	2.9699999999999998
1	Luís	Gonçalves	Audioslave	1.98
1	Luís	Gonçalves	Alanis Morissette	1.98
1	Luís	Gonçalves	Black Sabbath	0.99
1	Luís	Gonçalves	Apocalyptica	0.99
1	Luís	Gonçalves	Caetano Veloso	0.99
2	Leonie	Kähler	Audioslave	13.860000000000001
2	Leonie	Kähler	Alice In Chains	0.99
2	Leonie	Kähler	AC/DC	0.99
2	Leonie	Kähler	Buddy Guy	0.99
2	Leonie	Kähler	Chris Rea	0.99

2. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared return all Genres?



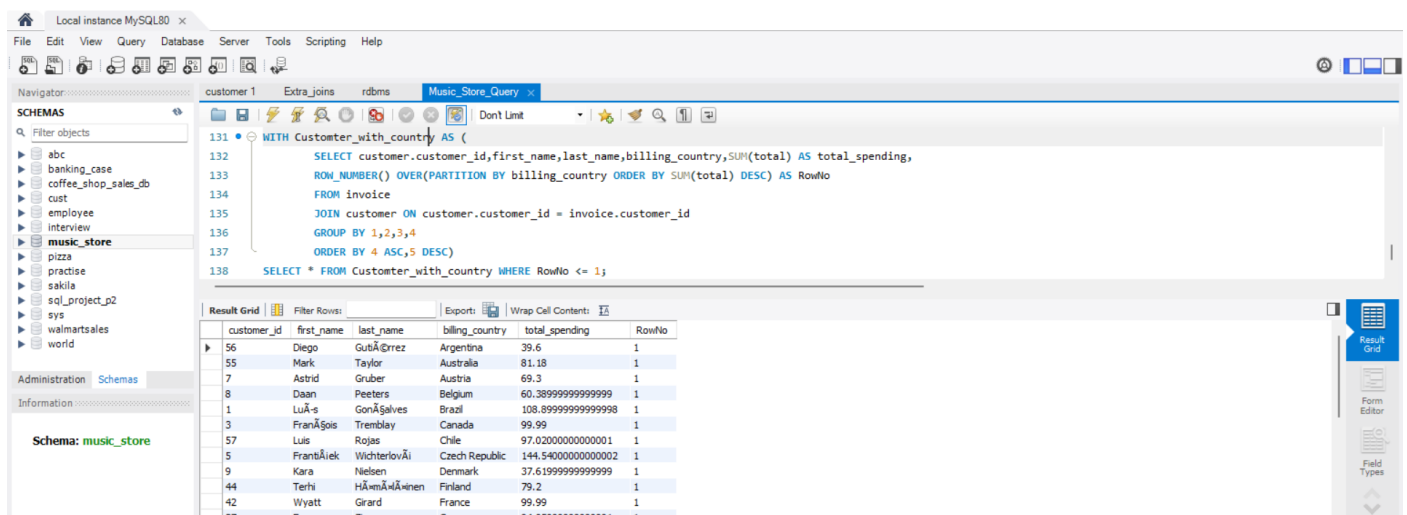
The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
WITH popular_genre AS (
  SELECT COUNT(invoice_line.quantity) AS purchases, customer.country, genre.name, genre.genre_id,
  ROW_NUMBER() OVER(PARTITION BY customer.country ORDER BY COUNT(invoice_line.quantity) DESC) AS RowNo
  FROM invoice_line
  JOIN invoice ON invoice_line.invoice_id = invoice.invoice_id
  JOIN customer ON customer.customer_id = invoice.customer_id
  JOIN track ON track.track_id = invoice_line.track_id
)
```

The result grid displays the following data:

purchases	country	name	genre_id	RowNo
1	Argentina	Rock	1	1
18	Australia	Rock	1	1
6	Austria	Rock	1	1
5	Belgium	Rock	1	1
26	Brazil	Rock	1	1
57	Canada	Rock	1	1
7	Chile	Rock	1	1
14	Czech Republic	Rock	1	1
6	Denmark	Rock	1	1
6	Finland	Rock	1	1
26	France	Rock	1	1
26	Germany	Rock	1	1

3. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount?



The screenshot shows the MySQL Workbench interface with a query window titled 'Music_Store_Query'. The query is as follows:

```
WITH Customer_with_country AS (
  SELECT customer.customer_id, first_name, last_name, billing_country, SUM(total) AS total_spending,
  ROW_NUMBER() OVER(PARTITION BY billing_country ORDER BY SUM(total) DESC) AS RowNo
  FROM invoice
  JOIN customer ON customer.customer_id = invoice.customer_id
  GROUP BY 1,2,3,4
  ORDER BY 4 ASC, 5 DESC
)
SELECT * FROM Customer_with_country WHERE RowNo <= 1;
```

The result grid displays the following data:

customer_id	first_name	last_name	billing_country	total_spending	RowNo
56	Diego	Gutiérrez	Argentina	39.6	1
55	Mark	Taylor	Australia	81.18	1
7	Astrid	Gruber	Austria	69.3	1
8	Daan	Peeters	Belgium	60.38999999999999	1
1	Luís	Gonçalves	Brazil	108.89999999999998	1
3	François	Tremblay	Canada	99.99	1
57	Luis	Rojas	Chile	97.02000000000001	1
5	František	Wichterlov	Czech Republic	144.94000000000002	1
9	Kari	Nielsen	Denmark	37.61999999999999	1
44	Terhi	Hämäläinen	Finland	79.2	1
42	Wyatt	Gard	France	99.99	1
37	Finn	Zimmermann	Germany	94.05000000000001	1